AgentGuard

Know Your Agent, built into the XRP Ledger

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Built with: XRP Ledger · Mastercard Agent Pay for Machines · AWS Bedrock AgentCore · Ledger · Quicknode
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> **Problem statement**
>
> AI agents are starting to pay for the data and services they use, but businesses have no way to verify which agent is paying, or to stop a rogue one everywhere at once.
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> **AgentGuard** gives every AI agent a passport on the XRP Ledger. Merchants accept only verified agents, and the agent can't spend a cent without AgentGuard's co-signature, which it gives only after checking the purchase against the agent's task. An instant kill switch freezes a rogue agent's spending everywhere, attacker included, and revokes its passport across every merchant in seconds.

1. The problem, in simple words

AI agents are no longer just chatbots. They research, book, buy data and call other services on their own, and more and more of those services charge per use. So agents increasingly **pay for things by themselves**.

That creates two problems nobody has solved well:

- **Merchants can't tell who an agent is.** A payment arrives from an AI agent. Is it a company's legitimate agent, or a scam bot?
- **Companies can't stop a rogue agent everywhere at once.** If an agent is tricked, hijacked or goes off-track, there is no single switch that stops it paying every merchant it deals with, let alone an attacker it has been lured to.

Today the choice is all or nothing: give the agent full access and hope, or approve every tiny payment by hand, which defeats the point of having an agent.

2. Our solution, in simple words

> **Think of an employee ID badge.** A company gives its employee a badge. Shops in the building only serve people wearing a valid badge. If the employee goes rogue, security deactivates the badge, and instantly no shop will serve them.

AgentGuard does the same for AI agents:

In the analogy	In AgentGuard
The employee	The AI agent
The ID badge	A digital **passport** stored on the XRP Ledger: "verified agent, owned by Acme, small purchases only, valid today"
Shops that only serve badge holders	**Merchants** (paid data services) that only accept payments from agents with a valid passport
A manager who must countersign every purchase	AgentGuard, which must **co-sign** every payment and only does so after checking: does this make sense for the agent's job?
Security deactivating the badge	The **kill switch**: AgentGuard stops co-signing, so the agent can't pay anyone, and the passport is revoked so no merchant will accept it

The key difference from a real badge: **the shops don't have to check the badge themselves.** On the XRP Ledger, the network automatically refuses a payment from an agent without a valid passport.

3. The use case: Acme's treasury analyst agent

The company. Acme Corp's treasury team holds money in euros, Japanese yen and XRP. Every morning they need a risk report: how much their exposure moved overnight, and what news could affect it.

The agent. Acme's AI Treasury Analyst holds a passport (small purchases only, valid for today's task). Its job: produce today's FX and crypto risk report. To do it, it buys data from pay-per-call vendors, paying in XRP directly on the XRP Ledger, with the news vendor paid in RLUSD (a US dollar stablecoin) through Mastercard's Agent Pay for Machines platform. Its wallet holds only a small daily float (10 XRP), and every payment needs AgentGuard's co-signature.

Merchant	What it sells	Price per call
FX Rates Co.	Live EUR/USD and USD/JPY exchange rates (real public data)	0.05 XRP
Crypto Prices Co.	Live XRP and BTC prices (real public data)	0.05 XRP
News Sentiment Co.	Today's market headlines plus an AI risk score (paid through Mastercard AP4M)	0.25 RLUSD
"Premium Alpha Signals" (the attacker)	Claims to sell secret "insider trading signals"	**5 XRP** (100× normal)

What happens

1. The agent reads its task and works out which data it needs.
2. It pays FX Rates, Crypto Prices and News Sentiment automatically. AgentGuard checks and co-signs each payment in about a second. No human is involved.
3. If the news shows high volatility, it decides on its own to buy a deeper data pull.

4. It writes the morning risk report for the treasurer.
5. "Premium Alpha Signals" tries to lure the agent into buying insider tips. AgentGuard **refuses to co-sign**: the vendor isn't on Acme's approved list and its price is 100× normal. The checker AI adds why: off-task and a compliance red flag.
6. The attacker keeps pushing and the agent tries to pay it directly, a sign it is being manipulated. The XRP Ledger rejects that payment because it lacks AgentGuard's signature. AgentGuard **fires the kill switch**: it stops co-signing for this agent and closes its Mastercard payment channel. With one approval from Acme's approver key (a Ledger device in production), the passport is revoked and the remaining float returns to Acme's treasury. The agent can now pay no one, attacker included.
7. Once Acme has checked everything is safe, it re-issues a passport with one approval.

> **One-line pitch:** Acme's AI analyst pays its own data bills, until someone tries to manipulate it. Then AgentGuard freezes its spending everywhere, attacker included, and cuts it off from every merchant in seconds.

4. How it works

Layer	What it does	Enforced by
1. Passport	Acme issues the agent an on-ledger credential with an expiry ("today only") and a level label ("small purchases"). The agent accepts it. Acme approves this with its approver key (a Ledger device in production; a software key in the demo). The ledger enforces the expiry; the level is a label that AgentGuard and merchants read.	XRP Ledger Credentials
2. Verified merchants only	Merchants switch on Deposit Authorization and approve the passport type they trust. Payments from agents without a valid passport are refused.	XRP Ledger validators
3. Co-signing checker	The agent's wallet needs two signatures, the agent's and AgentGuard's; the agent can't sign alone. Before co-signing, hard rules run first: approved merchant, price within 3× the vendor's usual price, and the daily budget. Then a separate AI checks that the purchase fits the task, using only structured facts (task, merchant, category, price, price history), never the merchant's text or the agent's own explanation. Result: co-sign, hold for a human, or refuse.	XRP Ledger multi-signing + AgentGuard (AWS Bedrock)
4. Kill switch	After repeated failed checks, AgentGuard stops co-signing at once, so the agent can't pay anyone. Acme then confirms with its approver key (or presses the kill button): the passport is deleted, the remaining float returns to the treasury, and the Mastercard payment channel is closed.	XRP Ledger validators + Mastercard AP4M

Routine payments run **fully autonomously**, each co-signed by AgentGuard in about a second. Humans step in only for the unusual: a held purchase, issuing a passport, or a revocation.

5. What already exists

Verifying AI agents, often called **Know Your Agent (KYA)**, is one of the most active areas in AI payments. AgentGuard builds on this work rather than claiming it is brand new.

Existing solution	What it does	What it leaves open
Mastercard Agent Pay / Agent Pay for Machines	Credentialing, permissioning, transacting and settling for AI agents across cards, accounts and stablecoins (AP4M launched June 2026).	The public announcement does not describe how an agent is revoked. Operated by Mastercard.
Skyfire (KYAPay)	Signed digital IDs for agents that merchants verify before checkout.	Each merchant must integrate the tool and choose to check.
Coinbase agentic wallets	Spending limits, session caps and transaction screening inside the wallet.	Limits live in wallet software; merchants can't see or rely on them.
ERC-8004 (Ethereum)	Registries for agent identity and reputation (live since January 2026).	Identity is recorded, but payments aren't automatically blocked by it.
AWS AgentCore Payments	Session spending caps, hidden credentials and audit trails for agents paying via x402.	Works with Coinbase and Stripe wallets; no XRP Ledger integration announced.
XRPL Credentials	Native on-ledger ID badges, designed mainly for human KYC.	We found no project using them as an agent passport with a kill switch.

6. Our novelty

The building blocks exist. What is new is **how we combine them and apply them to AI agents.**

1. The blockchain itself blocks unverified agents

Elsewhere, merchants must install a vendor's tool and choose to check an agent's ID. With AgentGuard, a merchant flips one setting on its XRP Ledger account, and the ledger's validators refuse payments from agents without a valid passport. **No SDK, no vendor, no merchant-side code.**

2. One action cuts an agent off everywhere

When AgentGuard stops co-signing, the agent can't pay **anyone**, including an attacker that would never ask for a passport. Deleting **one** passport also shuts the agent out of **every** merchant that requires it, within seconds, and closes its Mastercard payment channel. Anyone can verify it happened on the public ledger.

3. AI judgment triggers network-level enforcement (our key novelty)

Existing controls check **how much** an agent spends and **what kind** of merchant it pays. None we found check whether a purchase **fits the agent's task**. AgentGuard's checker does, and because its signature is required for every payment, its decision is enforced by the ledger's validators, not by the agent's goodwill. Repeated failures freeze the agent and revoke its passport.

> **AI decides something is wrong → the blockchain enforces it everywhere.** We found no existing product that links AI behaviour checks to network-level enforcement.

4. Passports with levels and expiry, readable by anyone

Passports carry a **level** ("small purchases only") and an **expiry** ("valid for today's task"), so a forgotten agent can't keep spending: once a passport expires, the ledger rejects payments that present it. The level is a label rather than a ledger-enforced limit; AgentGuard and merchants read it, and actual spending caps come from AgentGuard's checks and the agent's small balance. Because it all lives on a public ledger, any merchant, auditor or regulator can check an agent's status without asking anyone.

How this relates to Mastercard

Mastercard's Agent Pay for Machines names **credentialing** and **permissioning** as two of its four pillars. AgentGuard implements both natively on the XRP Ledger and adds instant, AI-triggered revocation. It is **complementary**: AgentGuard governs the agents that pay through Mastercard's platform.

Honesty note: "not done before" means we did not find it in our research, not proof that it doesn't exist.

7. Why the XRP Ledger

- **Built-in features:** Credentials, Deposit Authorization and multi-signing are part of the ledger itself. Elsewhere we would need custom smart contracts.
- **Fast and cheap:** payments confirm in 3–5 seconds for a fraction of a cent, ideal for frequent, small agent payments.
- **Dollar payments:** RLUSD lets agents pay in US dollars. Our demo pays in XRP; switching a merchant to RLUSD needs only one extra trustline.
- **Public audit trail:** every passport, payment and revocation is visible on a public explorer.

8. How each partner fits

Partner	Role in AgentGuard
Mastercard	One vendor is paid through **Agent Pay for Machines**: signed vouchers, RLUSD settlement on the XRPL EVM sidechain, fully autonomous. AgentGuard signs vouchers only after its checks pass, and the kill switch closes the AP4M channel.
AWS	The treasury analyst agent and the checker AI are built with **Strands Agents** and run on **Amazon Bedrock AgentCore**.

Partner	Role in AgentGuard
Ledger	Designed so that issuing and revoking passports, approving held purchases and returning a frozen agent's funds require a physical confirmation on a **Ledger device**, so a hacked server can't issue itself a passport or block a revocation. In the demo, a software key fills the device's slot in the signer list; the Speculos emulator or a borrowed device can replace it with no other changes.
Quicknode	All XRP Ledger traffic runs through our **Quicknode** endpoint, and the dashboard's live feed of payments and revocations comes from Quicknode **Streams/webhooks**.
XRP Ledger	Passports (Credentials), verified-only merchants (Deposit Authorization), co-signed agent wallets (multi-signing) and the public audit trail.

9. Demo (3 minutes)

Time	What judges see
0:00–0:30	**The problem:** AI agents now pay for things, but nobody can verify them or stop them.
0:30–1:00	**Passport issued:** Acme approves the analyst's passport with its approver key. It appears on the public XRP Ledger explorer.
1:00–1:45	**Autonomous work:** the agent buys FX rates and crypto prices on the XRP Ledger, presenting its passport, and news through Mastercard's platform. AgentGuard co-signs every payment, with no human involved. The dashboard updates live and the risk report appears.
1:45–2:15	**The attack:** "Premium Alpha Signals" pushes insider tips at 100× the normal price. AgentGuard refuses to co-sign (unapproved vendor, 100× price) and the checker AI explains why.
2:15–2:45	**Kill switch:** the manipulated agent tries to pay the attacker directly, and the payment is **refused by the XRP Ledger itself** because AgentGuard didn't sign. Acme approves with its approver key: the passport is revoked, the float returns to the treasury and the Mastercard channel closes. A payment to FX Rates with the old passport is now refused too.
2:45–3:00	**Why it matters:** open, ledger-enforced trust for AI agents that any merchant can use with one setting.

10. Under the hood

Part	Built with
Passports, verified-only merchants, kill switch	XRP Ledger testnet: CredentialCreate / CredentialAccept / CredentialDelete, Deposit Authorization, DepositPreauth, payments presenting a credential, and multi-signing on the agent wallet (signer weights: agent 1, AgentGuard 1, Acme's approver key 2; quorum 2; master key disabled), so routine payments need agent + AgentGuard and Acme alone can revoke and return funds (Python, xrpl-py)
Agent payments	XRP Ledger testnet: XRP payments presenting the passport, multi-signed by the agent and AgentGuard (Python, xrpl-py). Mastercard lane: AP4M EIP-712 vouchers, RLUSD on XRPL EVM testnet (chain ID 1449000), x402 pay-per-call (Python, web3)
Treasury analyst agent and checker AI	Strands Agents on Amazon Bedrock AgentCore; hard rule checks in plain Python run before the AI step
Approvals	Demo: a software approver key in the Ledger device's signer slot. Target: a Ledger device (Device Management Kit XRP signer) or the Speculos emulator, with a device-signed multisig approval as fallback if credential transactions aren't supported yet
Merchants and dashboard	Python (FastAPI); live feed via Quicknode Streams/webhooks

11. Honest limitations

- **Two networks.** Ledger-level blocking and co-signing apply to payments on the main XRP Ledger, where the core demo runs. On Mastercard's rail (the EVM sidechain), enforcement comes from merchants checking the passport before serving, AgentGuard refusing to sign vouchers, and the kill switch closing the payment channel.
- **Merchants must opt in.** It's one setting, but adoption is still needed. Deposit Authorization also blocks a merchant's other payers unless they are approved too.
- **The checker AI can be wrong.** Uncertain purchases are held for a human, revocation only follows repeated failures, and a passport can be re-issued with one approval.
- **AgentGuard is a required signer.** If AgentGuard is down, the agent can't pay, which is the safe default. A hacked AgentGuard still can't move money alone, because every payment also needs the agent's signature, and the agent's wallet holds only a small daily float.
- **Passports are public.** They hold only the issuer and passport type, never personal data.
- **Passport levels are labels.** The ledger checks that a valid, unexpired passport exists, not how much each purchase costs. Spending limits come from AgentGuard's checks and the small balance the agent holds.
- **No Ledger device in the demo yet.** We don't have a Ledger device on site, so a software key stands in for it; swapping in a device or the Speculos emulator changes only which key signs. Credential transactions are recent XRP Ledger types and Ledger's XRP tooling is still pre-release, so if the device

can't sign them, approvals use a device-signed multisig transaction instead, and we report the gap to Ledger as developer feedback.

- **Prototype.** The demo runs on testnets. Production would need secure key storage and real data contracts.

12. Likely judge questions

"Doesn't Mastercard already do this?"

Mastercard sets the trust model for agent payments: credential, permission, transact, settle. AgentGuard implements the credential and permission pillars natively on the XRP Ledger, adds instant AI-triggered revocation, and governs agents that pay through Mastercard's own platform. It is complementary, not competing.

"On Mastercard's rail, isn't it just your software refusing?"

Partly, and we're upfront about it. The core flow runs on the main XRP Ledger, where validators enforce both the passport and AgentGuard's co-signature. On the EVM rail, merchants check the agent's passport on the XRP Ledger before serving, and revocation closes the agent's AP4M channel, so the agent loses its funded lane, not just our approval.

"Why would an agent pay for FX rates? They're free."

Real treasury teams pay for premium real-time data and analysis. Our vendors stand in for paid data providers; the data underneath is real.

"What if the checker AI makes a mistake?"

Uncertain cases go to a human. Revocation only happens after repeated failures or a human decision, and re-issuing takes one approval.

"What if the agent itself is hacked?"

The agent never holds spending power alone. Its wallet needs AgentGuard's co-signature, and the ledger rejects any payment signed by the agent alone, so a hijacked agent can't pay an attacker directly, even one that never checks passports. Once frozen, its remaining float goes back to Acme's treasury, and its Mastercard channel is closed.

"What if the checker itself gets prompt-injected?"

It never reads the merchant's text or the agent's explanation, which is where injected instructions would hide. Hard rules run first (approved merchant, price within 3× normal, daily budget), and the AI only sees structured facts: the task, merchant, category, price and price history. In the demo, the attacker's 5 XRP offer is 100× the usual 0.05 XRP, so a rule blocks it before the AI is even asked.

"Why would a merchant refuse money?"

Compliance. Regulated data vendors must know who is paying them under anti-money-laundering and sanctions rules. Deposit Authorization does that check automatically: only agents holding a passport from an issuer the merchant trusts can pay at all, so the merchant never has to screen an unknown payer by hand.

"Who issues passports, and what's the business model?"

The agent's owner, or a trusted verification provider (Mastercard's AP4M partners already include KYA providers such as t54 Labs). Merchants choose which issuers they trust. Issuers pay to issue and manage passports; merchants get protection for free with one setting.

Appendix: key terms

Term	Meaning
XRP Ledger (XRPL)	A public blockchain built for payments: 3–5 second confirmation, fractions of a cent per payment.
Wallet	An account on the ledger: a public address (like an account number) and a private secret (like a password).
Credential	A digital ID badge stored on the XRP Ledger, issued by one account to another, with a type and an optional expiry.
Deposit Authorization	An account setting: "only accept payments from senders I've approved", including holders of a specific credential.
Multi-signing	An account setting that requires several keys to approve a payment. Here, the agent's key and AgentGuard's key must both sign.
RLUSD	A US dollar stablecoin issued by Ripple.
Mastercard AP4M	Agent Pay for Machines: Mastercard's platform for AI agents paying per use, with funded payment channels and signed vouchers.
x402	A way for an API to charge per request: it replies "402: pay first", the agent pays, then gets the data.
Testnet	A practice version of a blockchain with free, worthless coins.

Sources

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- Mastercard Agent Pay for Machines press release
- Mastercard Agent Pay
- Know Your Agent overview (BlockEden)
- Amazon Bedrock AgentCore Payments (AWS blog)
- Ledger bounty page
- Quicknode XRPL documentation
- XRPL Agentic Transactions
- XAIP protocol